Response to the reviewer (third round)

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for the recommendation of minor revision and for pinpointing how the evidence should be characterised. We took both routes offered under Major comment 1. First, we quantified the uncertainty of the residual-tail correction and now report a combined lower bound. Second, we explicitly limited the full-model claim to the tested tail strata and the stated residual model. All new numbers come from the seeded scripts (function `tail_rows` in `make_figures.py`) and are stored in `results/tail_summary.json`. Section and table numbers refer to the revised manuscript.

Comment 1 (three-storey full-model claim). *The t_3-adjusted estimate (3.52) of the margin design has no confidence interval reflecting the uncertainty of the fitted tail model. Either quantify it and give a conservative lower bound, or limit the conclusion to the tested strata and the stated t_3 model.*

Response. We did both.

- *Quantified uncertainty.* The location and scale of the fitted $\ln(U_{full}/U_s)$ model are now resampled by a nonparametric bootstrap (4000 resamples of the 550 re-analysed log-ratios per design). Each bootstrap residual is added to a draw from the Jeffreys posterior of the sampled strata. For the margin design (8408 kg) this gives $\hat{\beta} = 3.52$, 95% interval [3.41, 3.61] and one-sided lower bound 3.43, conditional on the t_3 model. With the more extreme t_2 model the point estimate is 3.16. Table 6 now has these columns for all ten designs, kept separate from the sampled-strata intervals.
- *Limited claim.* Section 4.6 now gives two estimates of different standing: a statistical result on the sampled strata (3.61 [3.45, 3.73]) and a model-based engineering estimate for the complete full model (3.52, lower bound 3.43 under t_3). It states that the target is met with 95% confidence on the sampled strata. For the complete model it is met only under the t_3 residual model, which is not a distribution-free demonstration. The abstract and Conclusion 5 use the same distinction.

Comment 2 (drift-bias sensitivity in the headline findings). *Qualify in the abstract and discussion that RBDO-C95 meets the targets under the baseline surrogate, that the confidence bound addresses Monte Carlo sampling only, and that results near the serviceability threshold are sensitive to the B_1–B_2 drift error.*

Response.

- *Abstract.* It now says that the RBDO-C95 designs meet both targets with 95% confidence "in the baseline surrogate model". It adds that these bounds cover sampling uncertainty only and that drift results near the target remain sensitive to the 1% error of the B_1–B_2 drifts.
- *Methods and Results.* The same qualifier was added in Section 2.5: the confidence bound controls Monte Carlo sampling uncertainty and does not remove systematic approximation error. It

also appears in Section 4.6, in the Discussion ("Statistical confidence and acceptance") and in Conclusion 4, which names the nine-storey design as becoming inconclusive under the 1% drift bias.

- *Base frame.* The RBDO-C design of the base frame is now described as meeting the drift target only 0.007 above the threshold. A 1% drift bias would put it below (Section 4.2).

Comment 3 (model dependence of the residual-tail validation). *Explain why t_3 is a suitable conservative sensitivity case, and state that it is a stress-test assumption rather than a validated tail.*

Response. A paragraph "Why t_3, and what it is not" was added to Section 3.3. It states that t_3 is a stress-test assumption, not a validated distribution of the extreme tail. It also explains why it is conservative relative to the data. We compared the 5 500 standardised log-ratios of strata B and C of the ten designs with the two models:

Standardised log-ratio	Observed	Normal model	t_3 model
> 2	78	125	383
> 3	10	7.4	159
> 4	0	0.2	77

The largest observed value is 3.84. The data are close to normal up to about four standard deviations and far lighter-tailed than t_3. The model is therefore pessimistic within the observed range, while nothing is known beyond it. The paragraph ends by stating that the residual estimates are conditional on this assumption. The t_2 case was added to show how much the result depends on the tail model.

Comment 4 (scope of the novelty claim). *Keep the full-text versus abstract distinction and restrict claims to the seven frames and the stated assumptions.*

Response. The distinction between the two studies read in full and those assessed from abstracts only († in Table 1) is retained. The contribution statement still states that we make no claim about the latter. The scope sentence at the end of the contribution list now restricts what is demonstrated to the seven planar frame configurations, the component-based ULS and the stated probabilistic model.

Additional comments

- *Abstract and conclusions.* Both now separate the baseline-surrogate result, the sampled-strata full-model result and the residual-adjusted estimate for the three-storey frame (Conclusion 5 is new).
- *Confidence terminology.* "Meets the target with 95% confidence" is now used only where the stated lower bound exceeds the target. The model under which this holds is always named: the baseline surrogate model, the sampled strata, or the t_3 residual model.
- *Table 6.* It is restructured into three groups: stratum C, sampled strata only (statistical intervals), and sampled strata plus the t-model residual (model-conditional intervals, with the bootstrap). The caption explains the difference.
- *Table 16.* The caption now flags the results that become below or inconclusive under the 1% drift perturbation, and it states that the lower bounds address sampling uncertainty only.
- *Table 17.* The caption states that RBDO-C-MU was optimised for two frame configurations only, and that c.o.v. 0.10 was not optimised.

- *Author information.* The author and affiliation placeholders remain in this version. They must be completed by the authors before formal submission.
- *Reproducibility.* The manuscript names the commit that produced its results. A release tag and DOI will be created on acceptance.